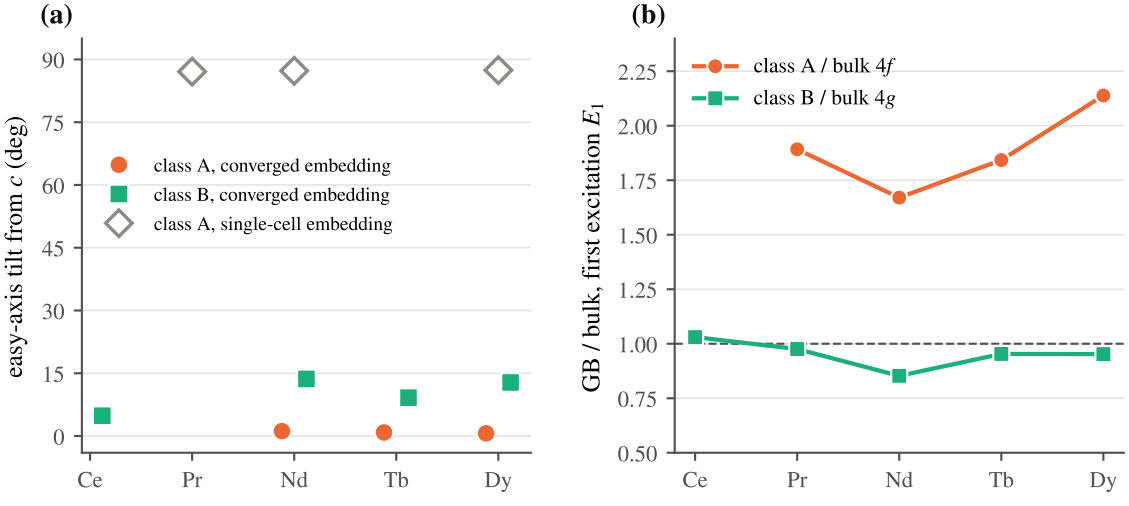
(a)
easy-axis tilt from c (deg)
90
75
60
45
30
15
0
class A, converged embedding
class B, converged embedding
class A, single-cell embedding
Ce
Pr
Nd
Tb
Dy
(b)
GB / bulk, first excitation E_1
2.25
2.00
1.75
1.50
1.25
1.00
0.75
0.50
class A / bulk 4f
class B / bulk 4g
Ce
Pr
Nd
Tb
Dy